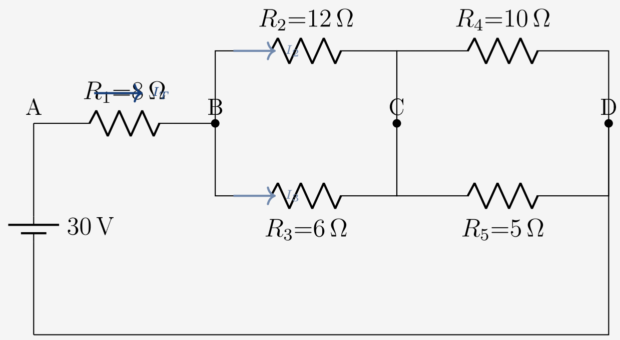
Circuitos de Corriente Continua

5 Ejercicios Resueltos — Circuitos Mixtos

$I_T \cdot V$ por tramos · P por bloques · Pilas en serie · Resistencias mixtas

Ejercicio 1

$R_1 = 8\,\Omega$, $R_2 = 12\,\Omega$, $R_3 = 6\,\Omega$, $R_4 = 10\,\Omega$, $R_5 = 5\,\Omega$, $\varepsilon = 30\,\text{V}$.
Topologia: R_1 – $[R_2 \| R_3]$ – $[R_4 \| R_5]$



Solucion

a) Resistencia equivalente

$$R_{23} = \frac{12 \times 6}{12 + 6} = \frac{72}{18} = 4\,\Omega \qquad R_{45} = \frac{10 \times 5}{10 + 5} = \frac{50}{15} = \frac{10}{3} \approx 3{,}33\,\Omega$$

$$\boxed{R_{eq} = R_1 + R_{23} + R_{45} = 8 + 4 + 3{,}33 = 15{,}33\,\Omega}$$

b) Intensidad total

$$I_T = \frac{30}{15{,}33} \approx \boxed{1{,}957\,\text{A}}$$

c) Voltaje en cada tramo

$$V_{R1} = I_T{\cdot}R_1 = 1{,}957{\times}8 \approx \mathbf{15{,}66}\,\text{V} \qquad V_{BC} = 1{,}957{\times}4 \approx \mathbf{7{,}83}\,\text{V} \qquad V_{CD} = 1{,}957{\times}3{,}33 \approx \mathbf{6{,}51}\,\text{V}$$

$$\text{Verif.: } 15{,}66 + 7{,}83 + 6{,}51 = 30\,\text{V} \checkmark$$

d) Intensidades en los paralelos

$$I_{R2} = \frac{7{,}83}{12} \approx 0{,}653\,\text{A} \quad I_{R3} = \frac{7{,}83}{6} \approx 1{,}305\,\text{A} \quad I_{R4} = \frac{6{,}51}{10} \approx 0{,}651\,\text{A} \quad I_{R5} = \frac{6{,}51}{5} \approx 1{,}302\,\text{A}$$

e) Potencia en el bloque B–C y bloque C–D

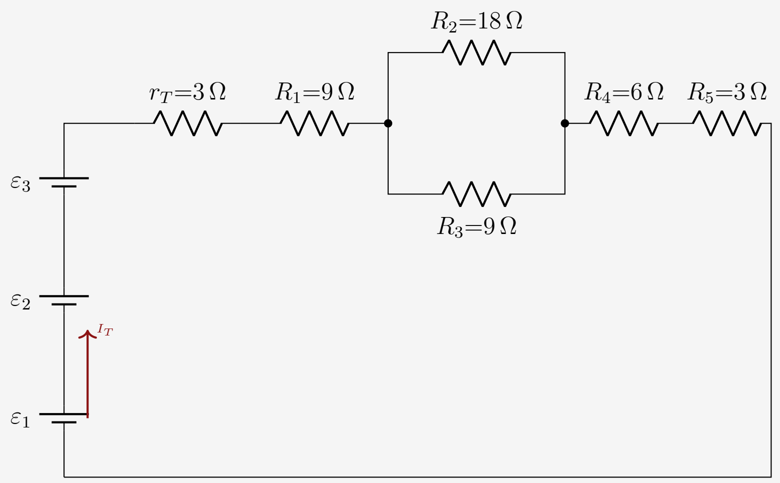
$$P_{BC} = V_{BC} \cdot I_T = 7{,}83 \times 1{,}957 \approx \boxed{15{,}32\,\text{W}} \qquad P_{CD} = 6{,}51 \times 1{,}957 \approx \boxed{12{,}74\,\text{W}}$$

f) Potencia total

$$P_{total} = \varepsilon \cdot I_T = 30 \times 1{,}957 \approx \boxed{58{,}7\,\text{W}}$$

Ejercicio 2

Tres pilas en serie alimentan el circuito.
$\varepsilon_1 = 6\,\mathrm{V}$, $\varepsilon_2 = 6\,\mathrm{V}$, $\varepsilon_3 = 6\,\mathrm{V}$, $r_1 = r_2 = r_3 = 1\,\Omega$.
$R_1 = 9\,\Omega$, $R_2 = 18\,\Omega$, $R_3 = 9\,\Omega$, $R_4 = 6\,\Omega$, $R_5 = 3\,\Omega$.
Topologia: R_1 – $[R_2 \| R_3]$ – R_4 – R_5



Solucion

a) FEM total y resistencia interna

$$\varepsilon_T = 6 + 6 + 6 = \boxed{18\,\mathrm{V}} \qquad r_T = 1 + 1 + 1 = 3\,\Omega$$

b) Resistencia equivalente

$$R_{23} = \frac{18 \times 9}{18 + 9} = \frac{162}{27} = 6\,\Omega$$

$$R_{ext} = R_1 + R_{23} + R_4 + R_5 = 9 + 6 + 6 + 3 = 24\,\Omega \qquad R_{tot} = r_T + R_{ext} = 3 + 24 = \boxed{27\,\Omega}$$

c) Intensidad total

$$I_T = \frac{18}{27} = \boxed{0{,}667\,\mathrm{A}}$$

d) Tension terminal de la bateria

$$V_{bat} = 18 - 0{,}667 \times 3 \approx \boxed{16\,\mathrm{V}}$$

e) Voltaje en cada tramo

$$V_{R1} = 0{,}667{\times}9 \approx 6\,\mathrm{V} \quad V_{BC} = 0{,}667{\times}6 \approx 4\,\mathrm{V} \quad V_{R4} = 0{,}667{\times}6 \approx 4\,\mathrm{V} \quad V_{R5} = 0{,}667{\times}3 \approx 2\,\mathrm{V}$$

$$\text{Verif.: } 6 + 4 + 4 + 2 = 16\,\mathrm{V} = V_{bat}\ \checkmark$$

f) Intensidades en el paralelo B–C

$$I_{R2} = \frac{4}{18} \approx 0{,}222\,\mathrm{A} \qquad I_{R3} = \frac{4}{9} \approx 0{,}444\,\mathrm{A} \qquad 0{,}222 + 0{,}444 \approx 0{,}666 \approx I_T\ \checkmark$$

g) Potencia en el bloque B–C y en R_4

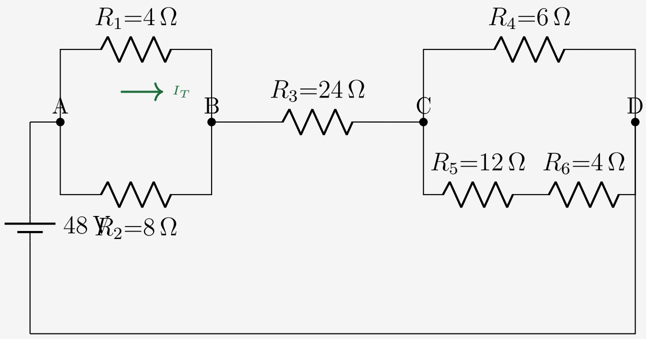
$$P_{BC} = V_{BC} \cdot I_T = 4 \times 0{,}667 \approx \boxed{2{,}67\,\mathrm{W}} \qquad P_{R4} = I_T^2 \cdot R_4 = 0{,}667^2 \times 6 \approx \boxed{2{,}67\,\mathrm{W}}$$

h) Potencia total

$$P_{total} = \varepsilon_T \cdot I_T = 18 \times 0{,}667 \approx \boxed{12\,\mathrm{W}}$$

Ejercicio 3

$\varepsilon = 48\,\mathrm{V}$, $R_1 = 4\,\Omega$, $R_2 = 8\,\Omega$, $R_3 = 24\,\Omega$, $R_4 = 6\,\Omega$, $R_5 = 12\,\Omega$, $R_6 = 4\,\Omega$.
Topologia: $[R_1 \| R_2]$ – R_3 – $[R_4 \| (R_5 + R_6)]$



Solucion

a) Resistencia equivalente

$$R_{12} = \frac{4 \times 8}{4 + 8} = \frac{32}{12} = \frac{8}{3} \approx 2{,}67\,\Omega$$

$$R_{56} = R_5 + R_6 = 12 + 4 = 16\,\Omega \qquad R_{CD} = \frac{R_4 \cdot R_{56}}{R_4 + R_{56}} = \frac{6 \times 16}{22} = \frac{96}{22} \approx 4{,}36\,\Omega$$

$$\boxed{R_{eq} = R_{12} + R_3 + R_{CD} = 2{,}67 + 24 + 4{,}36 = 31{,}03\,\Omega}$$

b) Intensidad total

$$I_T = \frac{48}{31{,}03} \approx \boxed{1{,}547\,\mathrm{A}}$$

c) Voltaje en el bloque A–B (V_{AB})

$$V_{AB} = I_T \cdot R_{12} = 1{,}547 \times 2{,}67 \approx \boxed{4{,}13\,\mathrm{V}}$$

$$I_{R1} = \frac{4{,}13}{4} \approx 1{,}033\,\mathrm{A} \qquad I_{R2} = \frac{4{,}13}{8} \approx 0{,}516\,\mathrm{A} \quad 1{,}033 + 0{,}516 \approx 1{,}549 \checkmark$$

d) Voltaje en R_3 (tramo B–C)

$$V_{R3} = I_T \cdot R_3 = 1{,}547 \times 24 \approx \boxed{37{,}13\,\mathrm{V}}$$

e) Voltaje en el bloque C–D (V_{CD})

$$V_{CD} = I_T \cdot R_{CD} = 1{,}547 \times 4{,}36 \approx \boxed{6{,}74\,\mathrm{V}}$$

$$I_{R4} = \frac{6{,}74}{6} \approx 1{,}123\,\mathrm{A} \qquad I_{56} = \frac{6{,}74}{16} \approx 0{,}421\,\mathrm{A} \quad 1{,}123 + 0{,}421 \approx 1{,}544 \approx I_T \checkmark$$

Voltaje en R_5 y R_6 por separado:

$$V_{R5} = 0{,}421 \times 12 \approx 5{,}05\,\mathrm{V} \qquad V_{R6} = 0{,}421 \times 4 \approx 1{,}68\,\mathrm{V} \quad 5{,}05 + 1{,}68 \approx 6{,}73 \approx V_{CD} \checkmark$$

f) Potencia en R_3 y en el bloque C–D

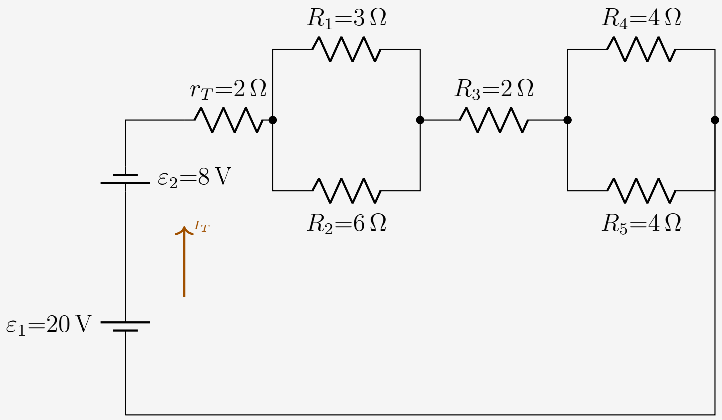
$$P_{R3} = I_T^2 \cdot R_3 = 1{,}547^2 \times 24 \approx \boxed{57{,}4\,\mathrm{W}} \qquad P_{CD} = V_{CD} \cdot I_T = 6{,}74 \times 1{,}547 \approx \boxed{10{,}42\,\mathrm{W}}$$

g) Potencia total

$$P_{total} = \varepsilon \cdot I_T = 48 \times 1{,}547 \approx \boxed{74{,}3\,\mathrm{W}}$$

Ejercicio 4

Dos pilas en serie (sentidos contrarios — una opuesta).
$\varepsilon_1 = 20\,\mathrm{V}$, $\varepsilon_2 = 8\,\mathrm{V}$ (sentido contrario), $r_1 = 1\,\Omega$, $r_2 = 1\,\Omega$.
$R_1 = 3\,\Omega$, $R_2 = 6\,\Omega$, $R_3 = 2\,\Omega$, $R_4 = 4\,\Omega$, $R_5 = 4\,\Omega$.
Topologia: $[R_1 \| R_2]$ – R_3 – $[R_4 \| R_5]$



Solucion

a) FEM resultante (pilas en sentidos contrarios, se restan)

$$\varepsilon_{net} = \varepsilon_1 - \varepsilon_2 = 20 - 8 = \boxed{12\,\mathrm{V}} \qquad r_T = r_1 + r_2 = 2\,\Omega$$

b) Resistencia equivalente

$$R_{12} = \frac{3 \times 6}{3 + 6} = 2\,\Omega \qquad R_{45} = \frac{4 \times 4}{4 + 4} = 2\,\Omega$$

$$R_{ext} = R_{12} + R_3 + R_{45} = 2 + 2 + 2 = 6\,\Omega \qquad R_{tot} = r_T + R_{ext} = 2 + 6 = \boxed{8\,\Omega}$$

c) Intensidad total

$$I_T = \frac{\varepsilon_{net}}{R_{tot}} = \frac{12}{8} = \boxed{1{,}5\,\mathrm{A}}$$

d) Tension terminal de la bateria equivalente

$$V_{bat} = \varepsilon_{net} - I_T \cdot r_T = 12 - 1{,}5 \times 2 = \boxed{9\,\mathrm{V}}$$

e) Voltaje por tramos

$$V_{12} = I_T \cdot R_{12} = 1{,}5 \times 2 = \mathbf{3}\,\mathrm{V} \qquad V_{R3} = 1{,}5 \times 2 = \mathbf{3}\,\mathrm{V} \qquad V_{45} = 1{,}5 \times 2 = \mathbf{3}\,\mathrm{V}$$

$$\text{Verif.: } 3 + 3 + 3 = 9\,\mathrm{V} = V_{bat} \checkmark$$

f) Intensidades en los paralelos

$$I_{R1} = \frac{3}{3} = 1\,\mathrm{A} \quad I_{R2} = \frac{3}{6} = 0{,}5\,\mathrm{A} \quad I_{R4} = \frac{3}{4} = 0{,}75\,\mathrm{A} \quad I_{R5} = \frac{3}{4} = 0{,}75\,\mathrm{A}$$

g) Potencia en cada bloque paralelo

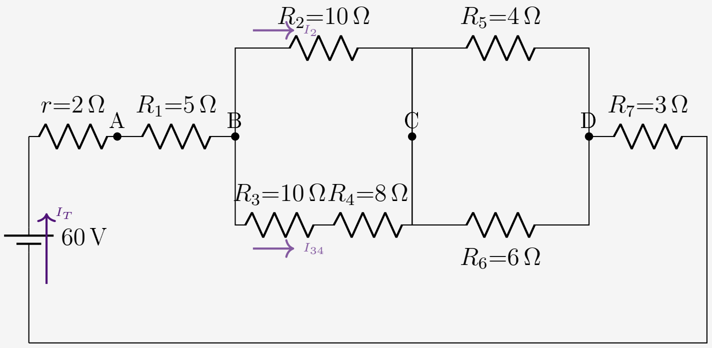
$$P_{12} = V_{12} \cdot I_T = 3 \times 1{,}5 = \boxed{4{,}5\,\mathrm{W}} \qquad P_{45} = 3 \times 1{,}5 = \boxed{4{,}5\,\mathrm{W}} \qquad P_{R3} = 1{,}5^2 \times 2 = \boxed{4{,}5\,\mathrm{W}}$$

h) Potencia total

$$P_{total} = \varepsilon_{net} \cdot I_T = 12 \times 1{,}5 = \boxed{18\,\mathrm{W}} \quad \Rightarrow \quad P_{12} + P_{R3} + P_{45} = 4{,}5 + 4{,}5 + 4{,}5 + P_{r_T} = 4{,}5 \times 3 + 1{,}5^2 \times 2 = 18\,\mathrm{W}\ \checkmark$$

Ejercicio 5

$\varepsilon = 60\,\text{V}$, $r = 2\,\Omega$.
$R_1 = 5\,\Omega$, $R_2 = 10\,\Omega$, $R_3 = 10\,\Omega$, $R_4 = 8\,\Omega$, $R_5 = 4\,\Omega$, $R_6 = 6\,\Omega$, $R_7 = 3\,\Omega$.
Topologia: R_1 – $[R_2 \| (R_3 + R_4)]$ – $[R_5 \| R_6]$ – R_7



Solucion

a) Resistencia equivalente

$$R_{34} = R_3 + R_4 = 10 + 8 = 18\,\Omega \qquad R_{BC} = \frac{R_2 \cdot R_{34}}{R_2 + R_{34}} = \frac{10 \times 18}{28} = \frac{180}{28} \approx 6{,}43\,\Omega$$

$$R_{CD} = \frac{R_5 \cdot R_6}{R_5 + R_6} = \frac{4 \times 6}{10} = 2{,}4\,\Omega$$

$$R_{ext} = R_1 + R_{BC} + R_{CD} + R_7 = 5 + 6{,}43 + 2{,}4 + 3 = 16{,}83\,\Omega \qquad R_{tot} = r + R_{ext} = 2 + 16{,}83 = \boxed{18{,}83\,\Omega}$$

b) Intensidad total

$$I_T = \frac{60}{18{,}83} \approx \boxed{3{,}186\,\text{A}}$$

c) Tension terminal y voltaje en R_1

$$V_{bat} = 60 - 3{,}186 \times 2 \approx 53{,}63\,\text{V} \qquad V_{R1} = 3{,}186 \times 5 \approx \mathbf{15{,}93}\,\text{V}$$

d) Voltaje en el bloque B–C (V_{BC})

$$V_{BC} = I_T \cdot R_{BC} = 3{,}186 \times 6{,}43 \approx \boxed{20{,}49\,\text{V}}$$

$$I_{R2} = \frac{20{,}49}{10} \approx 2{,}049\,\text{A} \qquad I_{34} = \frac{20{,}49}{18} \approx 1{,}138\,\text{A} \quad 2{,}049 + 1{,}138 \approx 3{,}187 \approx I_T\ \checkmark$$

Voltaje en R_3 y R_4 por separado:

$$V_{R3} = 1{,}138 \times 10 \approx 11{,}38\,\text{V} \qquad V_{R4} = 1{,}138 \times 8 \approx 9{,}10\,\text{V} \quad 11{,}38 + 9{,}10 \approx 20{,}48 \approx V_{BC}\ \checkmark$$

e) Voltaje en el bloque C–D (V_{CD})

$$V_{CD} = I_T \cdot R_{CD} = 3{,}186 \times 2{,}4 \approx \boxed{7{,}65\,\text{V}}$$

$$I_{R5} = \frac{7{,}65}{4} \approx 1{,}913\,\text{A} \qquad I_{R6} = \frac{7{,}65}{6} \approx 1{,}275\,\text{A} \quad 1{,}913 + 1{,}275 \approx 3{,}188 \approx I_T\ \checkmark$$

f) Potencia en el bloque B–C y bloque C–D

$$P_{BC} = V_{BC} \cdot I_T = 20{,}49 \times 3{,}186 \approx \boxed{65{,}3\,\mathrm{W}} \qquad P_{CD} = 7{,}65 \times 3{,}186 \approx \boxed{24{,}4\,\mathrm{W}}$$

g) Tabla resumen de potencias

Elemento	Formula	Potencia
r (int.)	$I_T^2 \cdot r = 3{,}186^2 \times 2$	$\approx 20{,}3\,\mathrm{W}$
R_1	$I_T^2 \cdot R_1 = 3{,}186^2 \times 5$	$\approx 50{,}7\,\mathrm{W}$
R_2	$I_{R2}^2 \cdot R_2 = 2{,}049^2 \times 10$	$\approx 42{,}0\,\mathrm{W}$
R_3	$I_{34}^2 \cdot R_3 = 1{,}138^2 \times 10$	$\approx 12{,}9\,\mathrm{W}$
R_4	$I_{34}^2 \cdot R_4 = 1{,}138^2 \times 8$	$\approx 10{,}4\,\mathrm{W}$
R_5	$I_{R5}^2 \cdot R_5 = 1{,}913^2 \times 4$	$\approx 14{,}6\,\mathrm{W}$
R_6	$I_{R6}^2 \cdot R_6 = 1{,}275^2 \times 6$	$\approx 9{,}7\,\mathrm{W}$
R_7	$I_T^2 \cdot R_7 = 3{,}186^2 \times 3$	$\approx 30{,}4\,\mathrm{W}$
TOTAL	$\varepsilon \cdot I_T = 60 \times 3{,}186$	$\approx \mathbf{191{,}2}\,\mathrm{W}$

$$20{,}3 + 50{,}7 + 42{,}0 + 12{,}9 + 10{,}4 + 14{,}6 + 9{,}7 + 30{,}4 = 191{,}0\,\mathrm{W} \approx P_{total}\ \checkmark$$